

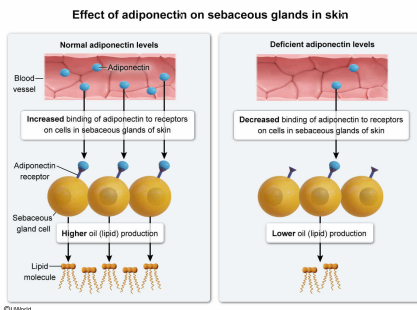
Researchers hypothesize that the hormone adiponectin significantly increases the activity of sebaceous glands in humans. Which of the following observations in patients deficient in adiponectin would best support this hypothesis?

- ☐ A. Decreased production of bile by the liver [2%]
- ✓ ☒ B. Decreased production of oils by the skin [93%]
- ☐ C. Decreased production of mucus by the trachea [2%]
- ☐ D. Decreased production of acid by the stomach [1%]

Correct

92% answered correctly

Explanation:



The skin contains several [structures](#), including sweat (sudoriferous) glands and sebaceous glands, both of which are [exocrine](#) glands located in the **dermis**:

- **Sweat glands** secrete sweat, a hypotonic solution (ie, a solution with a low solute concentration) that contributes to [thermoregulation](#). In warm environments, the evaporation of water from sweat on the skin leads to heat loss and cooling of the body. In addition, antimicrobial substances secreted by sweat glands can contribute to the [innate immune response](#).
- **Sebaceous glands** secrete sebum, an oil-containing (hydrophobic) substance that helps prevent water loss from the skin and hair. Sebum also contains certain substances (eg, [fatty acids](#)) that inhibit the survival of pathogens, contributing to innate immunity.

In this scenario, researchers hypothesize that the [hormone](#) adiponectin significantly *increases* sebaceous gland activity in humans. If this hypothesis is true, patients with adiponectin *deficiency* would be expected to exhibit **decreased** activity of the sebaceous (**oil**) glands in the skin. Therefore, observing **decreased production of oils in the skin** of these patients would **best support** this hypothesis.

(Choice A) Bile, which is synthesized and released by *liver cells*, is a *nonenzymatic* solution composed of bile salts, bile pigments (eg, bilirubin), and cholesterol. Bile salts have both a hydrophobic region that that can associate with lipid surfaces and a hydrophilic region that associates with water. Consequently, bile salts can act as detergents, breaking down large lipid globules into smaller droplets (micelles) via [emulsification](#).

(Choice C) Mucus—a thick, sticky, water-based substance containing salts, lipids, cellular debris, and protein—is produced by *specialized epithelial cells* in many areas of the body, including throughout the respiratory tract. Specifically, mucus coats the inside of the nose, pharynx, larynx, and [trachea](#) as well as the bronchi leading to the [lungs](#). This sticky mucus can [trap](#) foreign particles (eg, pathogens) to prevent them from reaching the lungs.

(Choice D) [Stomach](#) acid, which is secreted by [parietal cells](#), contains hydrochloric acid that helps to break down food and also maintains the low pH environment in the stomach that is necessary for activating the digestive enzyme pepsin. In addition, many pathogens cannot survive in the presence of stomach acid.

Educational objective:

Sebaceous glands are exocrine glands located in the dermis of the skin. Sebaceous glands secrete sebum, which is an oil-containing substance that prevents water loss from the skin and serves as a protective barrier against pathogens.

Time spent: 0 sec
Subject:
Biology

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Skin and Immune Systems